

2021 Trial Higher School Certificate PRELIMINARY EXAMINATION

Mathematics Standard

- Solutions including marking scale
- Mapping grid

We have endeavoured to ensure that the solutions are free of errors and follow the spirit of the syllabus in the methods used to solve the problems. However, individual teachers may opt for alternate solutions and/or may select a different marking system.

Marking Guidelines: Mathematics Standard, Preliminary HSC Trial Examination

1	B
2	B
3	D
4	B
5	C
6	D
7	B
8	A
9	D
10	D
11	A
12	A
13	B
14	C
15	B

ANSWERS

Question 16 (a)

Criteria	Marks
Provides correct solution	1

Medicare levy = $0.25(21\,890 - 18\,600) = \822.50

Question 16 (b)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

Mira's Medicare levy = 2.35% of $\$46\,678 = \$1\,096.93$

Difference = $\$1\,096.93 - 822.5 = \274.43

Question 17

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

Mean increases by 8 and the standard deviation remains the same

Question 18 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Arrives at Parramatta station $\rightarrow$ 12:45 pm + 20 mins $\rightarrow$ 1:05 pm

Catches 1:24 pm train to Auburn

Arrives at Auburn station 1:41 pm

Arrives at her office $\rightarrow$ 1:41 pm + 20 mins $\rightarrow$ 2:01 pm

Question 18 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

5.6 L $\rightarrow$ 100 km

5.6/100 L $\rightarrow$ 1 km

5.6/100(12)

i.e. 0.672 L

Cost = 0.672(1.45) = \$0.97

Question 19 (a)

Criteria	Marks
• Provides correct solution	1

Monthly pay = \$60 000/12 = \$5 000

Question 19 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Let x be Joe's sales

$5000 = 4000 + (0.1)50x$

$x = 200$

i.e. 200 books

Question 19 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

\$1.20 per 1000 pages

Therefore 60 cents for 500 pages

Each book printed Pitas receives 60 cents

\$60 000 = $0.6x$ where x is the number of books sold

$x = 100\,000$

Therefore 100 001 books

Question 20 (a)

Criteria	Marks
Provides correct solution	1

$$P(\text{Number less than 36}) = 35/50 = 7/10$$

Question 20 (b)

Criteria	Marks
Provides correct solution	1

$$P(\text{Not blue}) = 1 - P(\text{blue}) = 1 - 12/50 = 38/50 = 19/25$$

Question 20 (c)

Criteria	Marks
Provides correct solution	1

$$P(\text{either blue or yellow}) = 28/50 = 14/25$$

Question 20 (d)

Criteria	Marks
Provides correct solution	1

$$P(\text{divisible by 7}) = 7/50$$

$$P(\text{Not divisible by 7}) = 1 - 7/50 = 43/50$$

Question 21 (a)

Criteria	Marks
Provides correct solution	1

Median approx. 24

Question 21 (b)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$Q_u = 30$$

$$Q_L = 17.5$$

$$\text{IQR} = 30 - 17.5 = 12.5$$

Question 22

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$\text{Taxable income} = \$146\,000 - \$15\,000 = \$131\,000$$

$$\text{Tax payable on } \$131\,000 = \$17\,850 + 0.38(\$131\,000 - \$80\,000) = \$37\,230$$

Question 23 (a)

Criteria	Marks
Provides correct solution	1

$$AB^2 = (1.3 - 0.6)^2 + (19 - 11.1)^2 = 62.9$$

$$AB = 7.93$$

Question 23 (b)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$\text{Area} = 0.6 \times 19 + 0.5 \times 7.9 \times 0.7 = 14.165 \text{ m}^2$$

Question 23 (c)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$V = 14.165 \times 13.1 = 185.5615 \text{ m}^3 = 185.5615 \text{ KL} = 185561.5 \text{ L}$$

Question 24

Criteria	Marks
Provides correct solution	3
Majority of solution is correct	2
Provides some relevant information	1

Shower: 170L per 10 mins $\rightarrow$ 17 L per 1 min i.e. $17 \times 5 = 85 \text{ L}$ for 5 mins

Washing machine: $140 \times 3 = 420 \text{ L}$

Washing car: $2 \times 450 = 900 \text{ L}$

Total = $85 + 420 + 900 = 1405 \text{ L} = 1.405 \text{ kL}$

Total cost = $1.405 \times 1.85 = \$2.59925 = \2.60

Question 25 (a)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

Lower extreme = 80

$Q_L = 90$

$Q_u = 120$

Median = 100

Upper extreme = 130

Question 25 (b)

Criteria	Marks
Provides correct solution	1

50%

Question 26 (a)

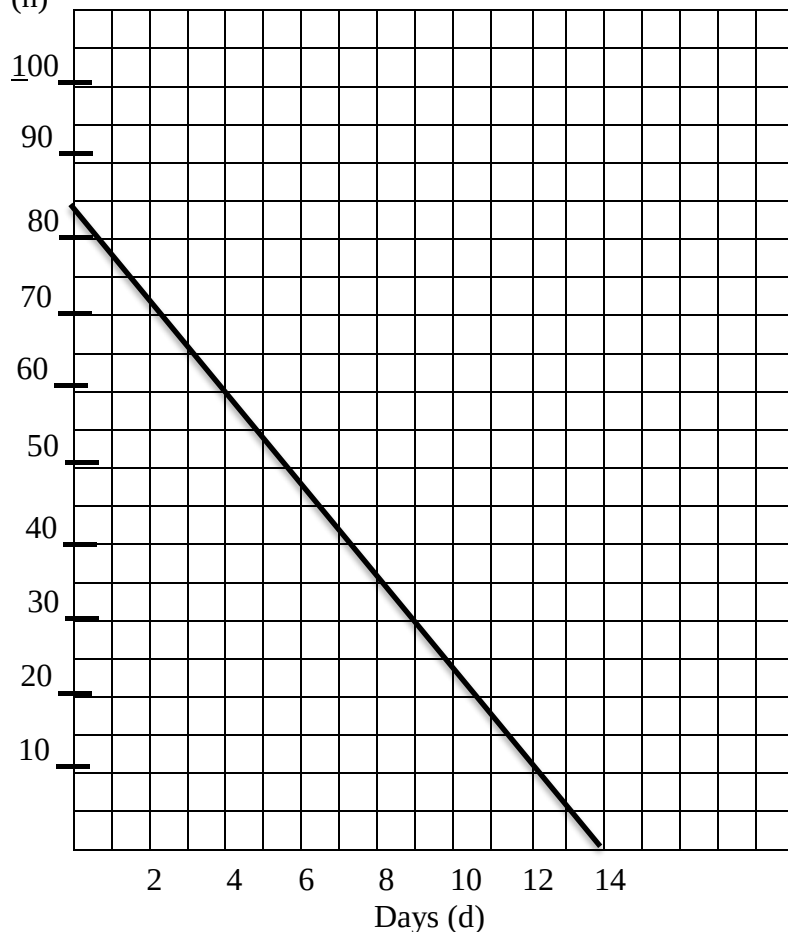
Criteria	Marks
Provides correct solution	1

d	0	2	4	6	8	10	12	14
n	84	72	60	48	36	24	12	0

Question 26 (b)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

Number of
infections
(n)



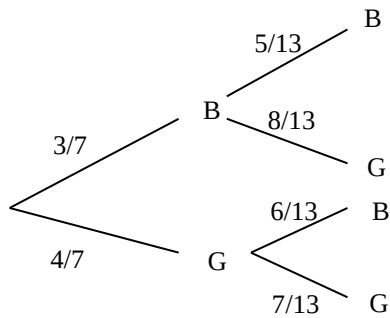
Question 26 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Gradient = -6. It is the rate of infections. It is decreasing at the rate of 6 per day.

Question 27 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

**Question 27 (b)**

Criteria	Marks
• Provides correct solution	1

$$P(\text{Both not blue}) = 1 - P(BB) = 1 - (3/7)(5/13) = 1 - 15/91 = 76/91$$

Question 27 (c)

Criteria	Marks
• Provides correct solution	1

$$P(\text{at least one green}) = 1 - P(GG) = 1 - (4/7)(7/13) = 1 - 4/13 = 9/13$$

Question 28 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\text{Range} = 36 - 2 = 34 \text{ mins} \quad \text{Mode} = 36 \text{ mins} \quad \text{Median} = 19 \text{ mins}$$

Question 28 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Let x be the missing number

$$24.625 = (371 + x)/16 \quad \text{i.e. } x = 23 \quad n = 3$$

Question 29

Criteria	Marks
• Provides correct solution	3
• Substitutes correctly into formula	2
• Provides some relevant information	1

$$A = 15/2(11 + 11) + 15/2(11 + 13) + 15/2(13 + 18) + 15/2(18 + 16)$$

$$A = 15/2(22) + 15/2(24) + 15/2(31) + 15/2(34)$$

$$A = 832.5 \text{ m}^2$$

Question 30 (a)

Criteria	Marks
• Provides correct solution	1

$$S = \$3\,800 - 14(5)$$

$$= \$3\,800 - 70$$

$$= \$3\,730$$

Question 30 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$3478 = 3800 - 14(n) \quad \text{where } n \text{ is number of years}$$

$$14n = 322$$

$$n = 23 \text{ yrs}$$

Question 30 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$110\% \text{ of pre tax cost} = 3800$$

$$1\% = 3800/110$$

$$10\% \text{ (GST)} = 3800/110 (10) = \$345.45$$

Question 31 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\text{Time} = 35\text{km}/(45\text{km/h}) = 0.777777.. \text{ hrs} = 46.6666... \text{ mins} = 47 \text{ mins}$$

Question 31 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\text{Speed} = 35/(58/60) = 36.21 \text{ minutes i.e. } 36 \text{ mins}$$

$$\text{Difference in time} = 47 - 36 = 11 \text{ mins}$$

Question 32 (a)

Criteria	Marks
• Provides correct solution	3
• Finds how far you can travel for \$1	2
• Provides some relevant information	1

$$14 \text{ Litres} \rightarrow 176 \text{ km}$$

$$1\text{L} \rightarrow \frac{176}{14} \text{ km}$$

$$\text{Cost} = \$1.46 \rightarrow 176/14 \text{ km}$$

$$\$1 \rightarrow \frac{\frac{176}{14}}{1.46} = 8.61 \text{ km}$$

$$\text{Now } \$66.6 = 8.61(66.6) = 573.426 = 573 \text{ km}$$

Question 32 (b)

Criteria	Marks
• Provides correct solution	1

$$70 \text{ km/h} = 70\,000/3\,600 \text{ m/s}$$

$$\text{Reaction distance} = 70\,000/3\,600 \times 1.8 = 35 \text{ m}$$

Question 32 (c)

Criteria	Marks
• Provides correct solution	1

$$\text{Stopping distance} = \text{reaction distance} + \text{braking distance} = 35 + 18.6 = 53.6 \text{ m}$$

Question 33 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\text{Thursday pay: } 3(13) + 3(14.60) + \frac{1}{2}(15.60) = \$90.60 \text{ plus uniform allowance of } 4.50(6.5) = \$29.25$$

$$\text{Thursday pay} = \$119.85$$

Question 33 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Saturday and Sunday pay: $2(1.5)(13.50) + 7(2)(13.50) = 229.50$

Total pay = $\$119.85 + 229.50 = \349.35

Question 34 (a)

Criteria	Marks
• Provides correct solution	1

Final value = $1590(1.216) = \$1\,933.44$

Question 34 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Interest compounded half yearly i.e interest is 2% for 6 months and period is 4 half yearly periods

$6700 = p(1.082)$ where p is the amount invested

$p = \$6192.24$

Question 35 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Principal = $12\,678 - 560 = 12\,118$

$I = 12\,118(0.135)(4) = \6543.72

Question 35 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Repayment = $(12\,118 + 6\,543.72)/48 = \388.79

Question 35 (c)

Criteria	Marks
• Provides correct solution	1

Amount paid = $388.79 \times 48 + 560 = \$19\,221.92$

Question 36 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

3 pm Wednesday (Lord Howe Island time) – 10.5 – 10 hrs = 3 pm Wed – 20.5 hr = 6:30 pm Tuesday (Cook Island time)

Question 36 (b)

Criteria	Marks
• Provides correct solution	3
• Majority of the calculations are correct	2
• Provides some relevant information	1

5:30 am Wed Alaska time + 19.5 hrs = 1 am Thursday (Lord Howe Island time)

Left Lord Howe Island = 1 am Thursday – (3 + 5 + 8) hrs = 1 am Thursday – 16 hrs = 9am Wednesday

Question 37 (a)

Criteria	Marks
• Provides correct solution	3
• Majority of the calculations are correct	2
• Provides some relevant information	1

$$\begin{aligned}
 SA &= 2(0.5)(25 + 47)20.8 + 27(22) + 47(22) + 22(25) + 22(27) + 2 \times \pi \times 10 \times 29 \\
 &= 1497.6 + 594 + 1034 + 550 + 594 + 1822.123 \\
 &= 6091.7 \text{ m}^2
 \end{aligned}$$

Question 37 (b)

Criteria	Marks
• Provides correct solution	1

$$6091.7/170 = 35.837 = 36 \text{ cans of paint}$$

Mathematics Standard

Preliminary HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes	Targeted Performance bands
1	1	MS-A1: Formulae and equations	MS11-1,6,9,10	1-2
2	1	MS-A2: Linear relationships	MS11-1,6,9,10	2-3
3	1	MS-S2: Relative frequency and probability	MS11-2,7,8,9,10	3-4
4	1	M1.1: Measurement and energy	MS11-3,4,9,10	2-3
5	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
6	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
7	1	M1.1: Measurement and energy	MS11-3,4,9,10	2-3
8	1	MS-M2: Working with time	MG11-1,6,9,10	3-4
9	1	F1.3: Budgeting and household expenses	MS11-2,5,6,9,10	2-3
10	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
11	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
12	1	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
13	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
14	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
15	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
16 (a)	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
16 (b)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
17	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
18 (a)	2	MS-M2: Working with time	MG11-1,6,9,10	2-3
18 (b)	2	MS-M2: Working with time	MG11-1,6,9,10	2-3
19 (a)	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
19 (b)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
19 (c)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
20 (a)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
20 (b)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
20 (c)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
20 (d)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
21 (a)	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
21 (b)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
22	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
23 (a)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
23 (b)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4

23 (c)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
24	3	F1.3: Budgeting and household expenses	MS11-2,5,6,9,10	2-3
25 (a)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
25 (b)	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
26 (a)	1	MS-A2: Linear relationships	MS11-1,6,9,10	2-3
26 (b)	2	MS-A2: Linear relationships	MS11-1,6,9,10	2-3
26 (c)	2	MS-A2: Linear relationships	MS11-1,6,9,10	2-3
27 (a)	2	MS-S2: Relative frequency and probability	MG11-1,6,9,10	2-3
27 (b)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	2-3
27 (c)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	2-3
28 (a)	2	S1.1: Classifying and representing data	MS11-2,7,8,9,10	2-3
28 (b)	2	S1.1: Classifying and representing data	MS11-2,7,8,9,10	2-3
29	3	M1.2: Perimeter, area and volume	MS11-3,4,9,10	2-3
30 (a)	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	3-4
30 (b)	2	F1.1: Interest and depreciation	MS11-2,5,6,9,10	3-4
30 (c)	2	F1.1: Interest and depreciation	MS11-2,5,6,9,10	3-4
31 (a)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
31 (b)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
32 (a)	3	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
32 (b)	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
32 (c)	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
33 (a)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
33 (b)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
34 (a)	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	2-3
34 (b)	2	F1.1: Interest and depreciation	MS11-2,5,6,9,10	4-5
35 (a)	2	F1.1: Interest and depreciation	MS11-2,5,6,9,10	4-5
35 (b)	2	F1.1: Interest and depreciation	MS11-2,5,6,9,10	4-5
35 (c)	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	4-5
36 (a)	2	MS-M2: Working with time	MG11-1,6,9,10	5-6
36 (b)	3	MS-M2: Working with time	MG11-1,6,9,10	5-6
37 (a)	3	M1.2: Perimeter, area and volume	MS11-3,4,9,10	5-6
37 (b)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	5-6